

MINI ERP + CRM OPERATIONS PORTAL

TECHNICAL ARCHITECTURE & SETUP GUIDE

Comprehensive Documentation Report — System Design, Deployment & Operations

1. How the Server Was Set Up

The backend service for the Mini ERP + CRM Operations Portal is architected as a modular, enterprise-grade Node.js & Express REST API using TypeScript. It enforces clean separation of concerns across routes, controllers, middleware, models, and utility layers.

Core Architecture & Technology Stack

- **Framework:** Express.js v4 with TypeScript (v5.3) for type safety and autocompletion.
- **Database Layer:** Prisma ORM (v5.10) with PostgreSQL for strict schema enforcement, automated migrations, and seed scripts.
- **Authentication:** JWT Access & Refresh token rotation with bcrypt password hashing (10 salt rounds).
- **Validation & Parsing:** Zod schema validation middleware for input sanitization and payload verification.
- **Logging System:** Pino structured JSON logger with request tracing (x-request-id correlation UUIDs).

Middleware & Security Pipeline

- **Helmet Security:** Configures HTTP response headers to protect against clickjacking, XSS, and MIME sniffing.
- **CORS Protection:** Strict origin allowlist validation enforcing allowed cross-origin web client requests.
- **Rate Limiter:** Express-rate-limit gating authentication endpoints to prevent brute-force attacks.
- **Audit Logging:** Asynchronous background logging capturing security access attempts (403 Forbidden) and critical CRUD operations.

```
# Database Connection (Neon Postgres / Local PostgreSQL)
DATABASE_URL="postgresql://user:pass@ep-hostname.aws.neon.tech/neondb?sslmode=require"

# JWT Security Secrets (Must be at least 32 characters long)
JWT_SECRET="prod-access-secret-key-min-32-chars-long!!"
JWT_REFRESH_SECRET="prod-refresh-secret-key-min-32-chars-long!!"
```

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ENVIRONMENT & EXECUTION GUIDE

```
# Application Configuration
Managing Credentials & Running Local Development
NODE_ENV="production"
PORT=10000
CORS_ORIGIN="https://mini-erp-crm-operations-portal.vercel.app"
```

```
# Frontend Environment Variable (Vite)
vite --mode=development --host 0.0.0.0 --port 3000 --base=/Mini-ERP-CRM-Operations-Portal/
cd Mini-ERP-CRM-Operations-Portal
```

2. How Environment Variables Are Managed

3. How to Run the Project Locally

```
# Start Backend Service (Runs on http://localhost:10000)
npm install && npm run dev

# In backend, dotenv is used to load environment variables. If any
# environment variables are strictly validated on server boot using zod schema in backend/src/config/env.ts. If any
# environment variable is missing, the application immediately throws an error with a descriptive message and instructions
# required to fix it. This is to ensure the application is deployed in a secure and reliable manner.
```

Step 2: Initialize Database & Seed Demo Data

Step 3: Demo Development Users

Email: admin@demo.com | Password: Demo@123

Email: sales@demo.com | Password: Demo@123

Email: warehouse@demo.com | Password: Demo@123

Email: accounts@demo.com | Password: Demo@123

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DEPLOYMENT STRATEGY & ASSUMPTIONS

Production Deployment Pipeline & Business Logic Specs

4. How to Deploy the Project

The project is pre-configured for deployment across Neon (PostgreSQL), Render (Web Service), and Vercel (Frontend SPA).

A. Database Deployment (Neon)

- **Create Project:** Log in to neon.tech, create a project named `mini-erp-db`, and select PostgreSQL.
- **Connection String:** Copy the pooled connection string (with `?sslmode=require`).
- **Run Migrations:** Execute ``npx prisma db push`` and ``npx prisma db seed`` against the Neon URL.

B. Backend Deployment (Render)

- **Service Type:** Create a new Web Service on render.com connected to the GitHub repository.
- **Build Command:** ``npm install && npx prisma generate && ./node_modules/.bin/tsc``
- **Start Command:** ``node dist/server.js``
- **Environment Keys:** Add `DATABASE_URL`, `JWT_SECRET`, `JWT_REFRESH_SECRET`, `CORS_ORIGIN`, and `NODE_ENV=production`.

C. Frontend Deployment (Vercel)

- **Import Repo:** Import repository into vercel.com with root directory set to ``frontend``.
- **Environment Variable:** Set ``VITE_API_BASE_URL`` to your Render backend URL (e.g. `https://...onrender.com/api/v1`).

5. Assumptions Made

During the implementation of the business domain requirements, the following technical and operational assumptions were established:

- **Challan Cancellation:** Reversing or cancelling a confirmed Sales Delivery Challan automatically restores all deducted line item quantities back to product stock inventory via stock movement ledger entries.
- **Price Snapshotting:** Unit prices are snapshot at the moment of Challan creation so future product price updates do not retroactively alter historic delivery invoices.
- **Lockout Policy:** User accounts lock out after 5 consecutive failed login attempts for 15 minutes to mitigate brute-force risks.
- **Challan Numbering:** Auto-generated sequentially using format ``CH-YYYY-XXXX`` (e.g., `CH-2026-0001`).
- **Warehouse Single Location:** Each product SKU is mapped to a primary warehouse storage location string (e.g. `Rack A-12`).